1. INTRODUCTION

Lung cancer is one of the most serious and life-threatening diseases worldwide, accounting for a significant number of cancer-related deaths each year. Early detection of lung cancer is crucial for increasing the chances of successful treatment and improving patient survival rates. Medical imaging techniques such as Computed Tomography (CT) scans are commonly used by healthcare professionals to identify abnormalities in lung tissues. However, manual analysis of CT scan images can be time-consuming and may be affected by human error, especially when dealing with large volumes of medical data.

Recent advancements in Artificial Intelligence (AI) and Deep Learning have enabled the development of intelligent systems capable of assisting medical experts in disease diagnosis. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated exceptional performance in image classification and medical image analysis tasks. These models can automatically learn complex patterns and features from images, making them suitable for detecting diseases from medical scans.

The proposed Lung Cancer Detection System utilizes the Xception deep learning architecture to classify lung CT scan images into three categories: Benign Cases, Malignant Cases, and Normal Cases. The system is trained using the IQ-OTHNCCD Lung Cancer Dataset and deployed through a Streamlit-based web application, allowing users to upload CT scan images and obtain instant predictions along with confidence scores. The developed system aims to provide a reliable, efficient, and user-friendly solution that can assist healthcare professionals in the early detection of lung cancer.

1.1 PROBLEM STATEMENT

Lung cancer remains one of the leading causes of cancer-related mortality worldwide. Accurate and early diagnosis is essential for effective treatment and improved patient outcomes. Traditional diagnostic procedures rely on the manual examination of CT scan images by radiologists, which can be time-consuming and subject to variations in interpretation.

With the increasing availability of medical imaging data, there is a growing need for automated systems that can assist in analyzing CT scan images quickly and accurately. Existing manual methods may not always provide consistent results, particularly when large numbers of scans need to be examined. Therefore, an intelligent computer-aided diagnostic system is required to support medical professionals by automatically classifying lung CT scan images into appropriate categories.

This project addresses the problem by developing a deep learning-based Lung Cancer Detection System that uses the Xception model to classify CT scan images into Benign, Malignant, and Normal categories, thereby supporting faster and more reliable diagnosis.

1.2 OBJECTIVE

The primary objective of this project is to develop an intelligent and automated Lung Cancer Detection System using deep learning techniques for the classification of lung CT scan images.

The specific objectives of the project are:

- To develop a deep learning model capable of accurately classifying lung CT scan images.
- To utilize the Xception transfer learning architecture for improved classification performance.
- To classify CT scan images into Benign Cases, Malignant Cases, and Normal Cases.
- To perform image preprocessing and data augmentation for enhancing model accuracy and generalization.
- To evaluate the performance of the trained model using appropriate metrics.
- To provide prediction results along with confidence scores and class probabilities.
- To deploy the trained model through a Streamlit-based web application for easy user interaction.
- To assist healthcare professionals by providing rapid preliminary diagnostic support.

1.3 SCOPE

The scope of this project is focused on the automated detection and classification of lung cancer from CT scan images using deep learning techniques. The system is designed to assist in the analysis of medical images by automatically identifying whether a CT scan belongs to a Benign, Malignant, or Normal category.

The project covers various stages including dataset preparation, image preprocessing, data augmentation, model training, performance evaluation, and deployment through a web-based interface. The developed application allows users to upload CT scan images and obtain real-time classification results.

The system is intended to function as a supportive diagnostic tool rather than a replacement for professional medical expertise. Future enhancements may include the use of larger datasets, integration with healthcare systems, support for additional disease categories, and the incorporation of more advanced deep learning models to improve diagnostic accuracy and usability.

1.4 ORGANISATION OF REPORT

This report is organized into seven major chapters that describe the design, development, implementation, and evaluation of the Lung Cancer Detection System.

Chapter 1 – Introduction presents the background of lung cancer diagnosis, the problem statement, objectives, scope, and an overview of the proposed lung cancer detection system.

Chapter 2 – Literature Survey discusses related research works, existing computer-aided diagnostic systems, medical image classification techniques, deep learning approaches, and the technologies used for the development of the proposed system.

Chapter 3 – System Analysis describes existing methods for lung cancer detection, their limitations, the proposed Xception-based classification system, its advantages, applications, and software requirement specifications.

Chapter 4 – System Design presents the UML diagrams and Data Flow Diagrams (DFDs) that illustrate the architecture, workflow, interactions, and data movement within the Lung Cancer Detection System.

Chapter 5 – Implementation explains the implementation of major modules including dataset preparation, image preprocessing, data augmentation, Xception model training, model evaluation, prediction generation, and Streamlit-based user interface deployment.

Chapter 6 – Testing presents the test cases, validation procedures, performance evaluation metrics, and testing outcomes used to verify the accuracy, functionality, and reliability of the developed system.

Chapter 7 – Results discusses the results obtained from the trained Xception model, including classification outputs, performance analysis, screenshots of the deployed application, and interpretation of the prediction results.

Finally, the report concludes with the project outcomes, future enhancement possibilities, references, appendix, and project mapping resources associated with the Lung Cancer Detection System.

2. LITERATURE SURVEY

2.1 RESEARCH PAPERS AND PUBLICATIONS

Table 2.1: Research papers related to Lung Cancer Detection uing Xception Learning

S.No	Title	Authors	Year	Outcomes
1	Lung Cancer Detection Using Deep Learning Techniques	A. Kumar, R. Sharma	2020	Developed a CNN-based model for lung cancer classification and achieved high accuracy in identifying cancerous lung nodules from CT scan images.
2	Automated Lung Cancer Detection from CT Images Using Transfer Learning	M. Al Mamun, S. Islam	2021	Applied transfer learning techniques to improve classification accuracy and reduce training time for lung cancer diagnosis.
3	Deep Learning-Based Computer-Aided Diagnosis System for Lung Cancer Detection	H. Wang, Y. Zhang	2020	Proposed a CAD system using CNN architectures that improved diagnostic efficiency and reduced manual intervention.
4	Lung Cancer Classification Using Xception Network and Medical Imaging	S. Gupta, P. Verma	2022	Utilized the Xception architecture for CT image classification and achieved improved performance compared to traditional CNN models.
5	Transfer Learning Approaches for Medical Image Classification	K. Rajesh, M. Prakash	2021	Demonstrated that pre-trained deep learning models significantly enhance medical image classification accuracy while reducing computational cost.
6	Artificial Intelligence in Lung Cancer Diagnosis: A Review	J. Singh, R. Kaur	2023	Reviewed AI-based techniques for lung cancer detection and highlighted the effectiveness of

				deep learning in medical image analysis.
7	CT Scan-Based Lung Disease Classification Using Deep Neural Networks	T. Ahmed, M. Hassan	2022	Developed a deep neural network model capable of accurately classifying various lung conditions from CT scan images.

The literature survey indicates that deep learning has become one of the most effective approaches for automated lung cancer detection and classification. Several studies have demonstrated that Convolutional Neural Networks and transfer learning models can successfully identify cancer-related patterns in CT scan images with high accuracy. Among various architectures, transfer learning models such as Xception have shown superior performance due to their ability to leverage pre-trained knowledge and extract complex image features efficiently. These research findings support the development of the proposed Lung Cancer Detection System, which utilizes the Xception architecture to classify lung CT scan images into Benign, Malignant, and Normal categories through an automated and user-friendly platform.

2.2 TECHNOLOGY

The Lung Cancer Detection System is developed using modern Deep Learning, Medical Image Analysis, and Web Application technologies that provide an efficient, accurate, and user-friendly platform for lung cancer classification. The system follows a deep learning-based architecture, where the trained Xception model performs image classification, while the Streamlit interface enables users to interact with the system through image uploads and result visualization.

1. **Xception:** Xception (Extreme Inception) is the core deep learning model used in this project. It is a powerful Convolutional Neural Network architecture that utilizes depthwise separable convolutions to improve feature extraction efficiency while reducing computational complexity. The Xception model is employed through transfer learning to classify lung CT scan images into Benign Cases, Malignant Cases, and Normal Cases. Its high accuracy and strong feature learning capabilities make it suitable for medical image classification tasks.

2. TensorFlow and Keras: TensorFlow and Keras are used for developing, training, and evaluating the deep learning model. These frameworks provide a comprehensive environment for building neural networks, performing transfer learning, handling image data, and saving trained models for deployment. TensorFlow also supports efficient model execution during prediction.

3. Scikit-learn: Scikit-learn is utilized for dataset splitting, model evaluation, and performance analysis. It provides functions for generating classification reports, confusion matrices, and validation metrics that help assess the effectiveness of the trained model.

4. Streamlit: Streamlit is used to develop the web-based user interface of the system. It enables users to upload lung CT scan images, perform predictions using the trained model, and visualize classification results through a simple and interactive interface. Streamlit simplifies deployment and provides an accessible platform for end users.

5. IQ-OTHNCCD Lung Cancer Dataset: The IQ-OTHNCCD Lung Cancer Dataset serves as the primary dataset for model training and evaluation. It contains CT scan images categorized into Benign Cases, Malignant Cases, and Normal Cases. The dataset provides sufficient image diversity to train the deep learning model effectively for lung cancer classification.

6. Visual Studio Code (VS Code): Visual Studio Code is used as the primary development environment for implementing the machine learning model and Streamlit application. It provides extensive support for Python programming, debugging, project management, and integration with deep learning frameworks.

The integration of these technologies enables the Lung Cancer Detection System to provide accurate CT scan image classification, efficient model deployment, reliable prediction generation, and a user-friendly platform for assisting in the early detection of lung cancer.

3. SYSTEM ANALYSIS

System analysis is the process of examining the existing methods, identifying their limitations, and proposing an improved solution to address the identified problems. In the context of lung cancer diagnosis, traditional methods often depend on manual interpretation of CT scan images, which can be time-consuming and prone to inconsistencies. The proposed Lung Cancer Detection System utilizes deep learning techniques to automate the classification process and improve diagnostic efficiency.

3.1 Existing System

Existing lung cancer detection systems primarily rely on manual examination of CT scan images by radiologists and medical experts. In these systems, healthcare professionals analyze CT scans to identify abnormalities, lung nodules, and cancerous tissues. While this approach is widely used in medical practice, the increasing volume of medical imaging data has made the diagnostic process more challenging and time-intensive.

Some computer-aided diagnosis systems have been developed to assist radiologists, but many traditional systems depend on handcrafted feature extraction techniques and conventional machine learning algorithms. These approaches often require extensive domain expertise and may not effectively capture complex patterns present in medical images.

As a result, the need for more intelligent and automated diagnostic systems has increased, leading to the adoption of deep learning-based approaches for medical image analysis.

3.1.1 Limitations of Existing System

- Manual analysis of CT scan images is time-consuming and labor-intensive.
- Diagnostic results may vary depending on the experience and expertise of the radiologist.
- Traditional machine learning approaches require manual feature extraction.
- Existing systems may struggle to identify subtle patterns present in complex medical images.

- Large volumes of imaging data can increase the workload of healthcare professionals.
- Conventional approaches may provide lower classification accuracy compared to deep learning models.
- Delays in diagnosis can affect treatment planning and patient outcomes.

3.2 Proposed System

The proposed Lung Cancer Detection System is an automated deep learning-based solution designed to classify lung CT scan images into three categories: Benign Cases, Malignant Cases, and Normal Cases. The system employs the Xception transfer learning architecture, which has proven highly effective for image classification tasks due to its powerful feature extraction capabilities.

The system performs image preprocessing and data augmentation to improve model generalization and classification accuracy. The trained model is integrated into a Streamlit-based web application that allows users to upload CT scan images and receive instant predictions along with confidence scores and class probabilities.

By automating the classification process, the proposed system aims to provide rapid and reliable preliminary diagnostic support while reducing the workload of healthcare professionals.

3.2.1 Advantages of Proposed System

- Provides automated classification of lung CT scan images.
- Reduces the time required for preliminary diagnosis.
- Utilizes the Xception transfer learning model for improved accuracy.
- Supports classification into Benign, Malignant, and Normal categories.
- Eliminates the need for manual feature extraction.
- Generates confidence scores and class probabilities for better interpretation.
- Offers a simple and user-friendly web interface through Streamlit.
- Improves consistency and reliability in image classification.
- Can assist healthcare professionals in decision-making.
- Easily scalable for future enhancements and larger datasets.

3.3 Applications

The Lung Cancer Detection System can be applied in various healthcare and medical imaging environments to support early diagnosis and clinical decision-making. Some of the major applications include:

- Hospitals and diagnostic centers for preliminary lung cancer screening.
- Radiology departments for assisting in CT scan image analysis.
- Medical research institutions conducting studies on lung diseases and cancer detection.
- Healthcare organizations seeking computer-aided diagnostic support systems.
- Educational institutions for research and learning in medical image analysis and deep learning.
- Telemedicine platforms requiring remote diagnostic assistance.
- Artificial intelligence research projects related to disease classification and medical imaging.
- Clinical decision support systems aimed at improving diagnostic efficiency and accuracy.

3.4 Software Requirements Specification

The Software Requirements Specification (SRS) defines the software environment and resources required for the successful development, training, deployment, and execution of the Lung Cancer Detection System.

Software Requirements:

Table 3.1: Software Requirements

Component	Specification
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.x
Deep Learning Framework	TensorFlow, Keras
Machine Learning Library	Scikit-learn
Web Framework	Streamlit

IDE	Visual Studio Code
Data Visualization	Matplotlib, Seaborn
Image Processing	Pillow (PIL), NumPy

Hardware Requirements:

Table 3.2: Hardware Requirements

Component	Specification
Processor	Intel Core i5 or higher
RAM	Minimum 8 GB
Storage	Minimum 10 GB Free Space
GPU (Optional)	NVIDIA GPU for faster training
Display	Standard Monitor
Internet Connection	Required for package installation and updates

This specification provides the necessary software and hardware environment for the successful execution and deployment of the Lung Cancer Detection System.

4. SYSTEM DESIGN

System Design is the process of defining the architecture, components, modules, interfaces, and data flow of a software system. It provides a blueprint for the implementation of the Lung Cancer Detection System by illustrating how different components interact with each other to perform lung cancer classification. The system design focuses on user interaction, medical image processing, deep learning-based classification, and result visualization. Various UML diagrams and Data Flow Diagrams (DFDs) are used to represent the structure, behavior, and data movement within the system.

4.1 UML Diagrams

Unified Modeling Language (UML) diagrams are used to visualize the design and functionality of the Lung Cancer Detection System. UML diagrams provide a graphical representation of the system's structure, workflow, and interactions between different components. They help in understanding the behavior of the system and serve as a foundation for implementation.

The UML diagrams used in this project are:

- UML Activity Diagram
- UML Use Case Diagram
- UML Class Diagram
- UML Sequence Diagram

These diagrams illustrate the workflow of lung cancer classification, user interactions, relationships among system components, and the sequence of operations performed during prediction generation.

4.1.1 Activity Diagram

The Activity Diagram represents the workflow of the system from CT scan image upload to prediction result generation. It illustrates the sequence of activities involved in preprocessing the image, loading the trained Xception model, performing classification, generating confidence scores, and displaying the final prediction to the user.

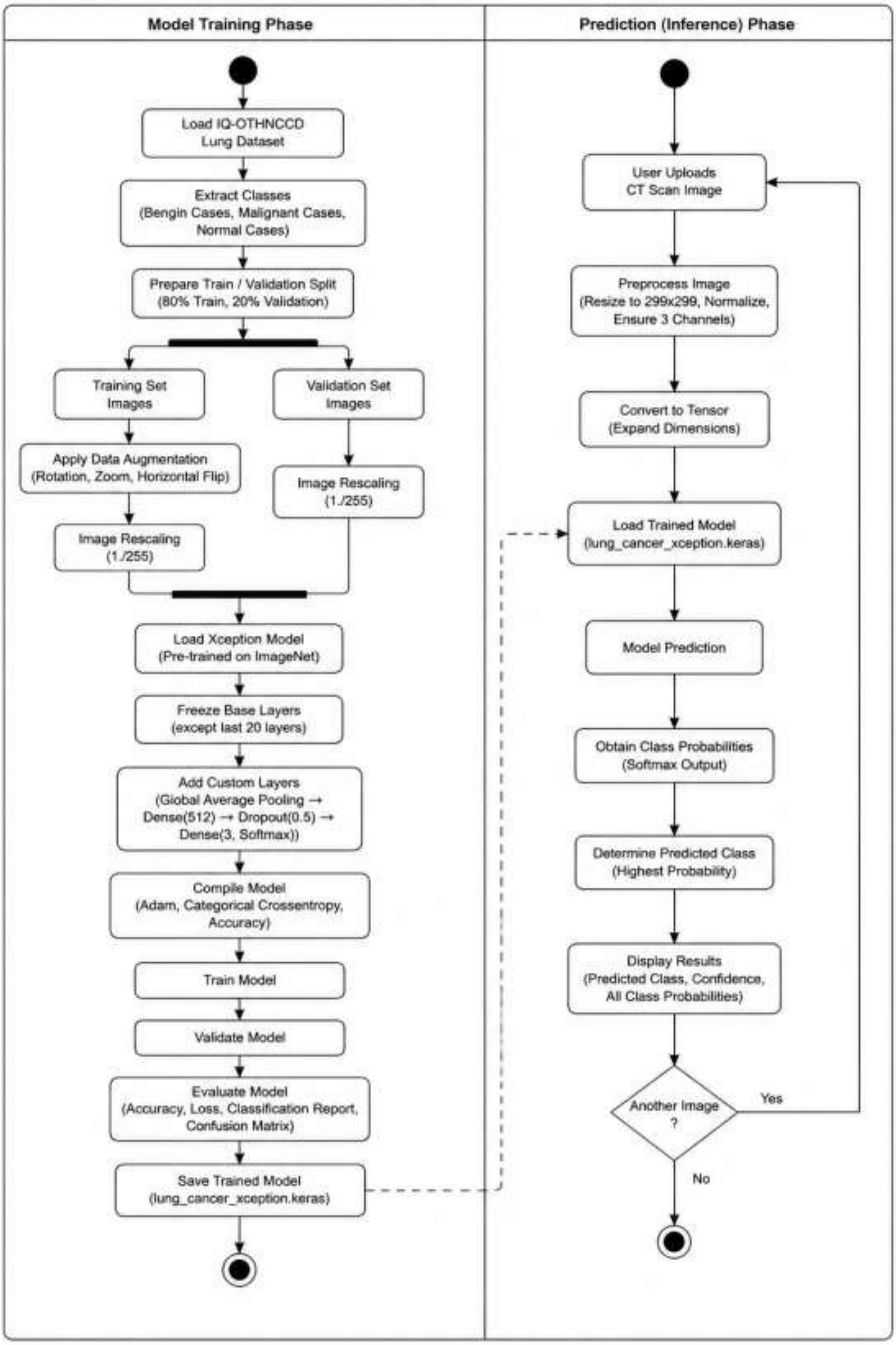


Figure 4.1: Activity Diagram

4.1.2 Use Case Diagram

The Use Case Diagram describes the interaction between the user, administrator, and the system. It identifies the functionalities provided by the application, including image upload, prediction generation, result visualization, model management, and report generation.

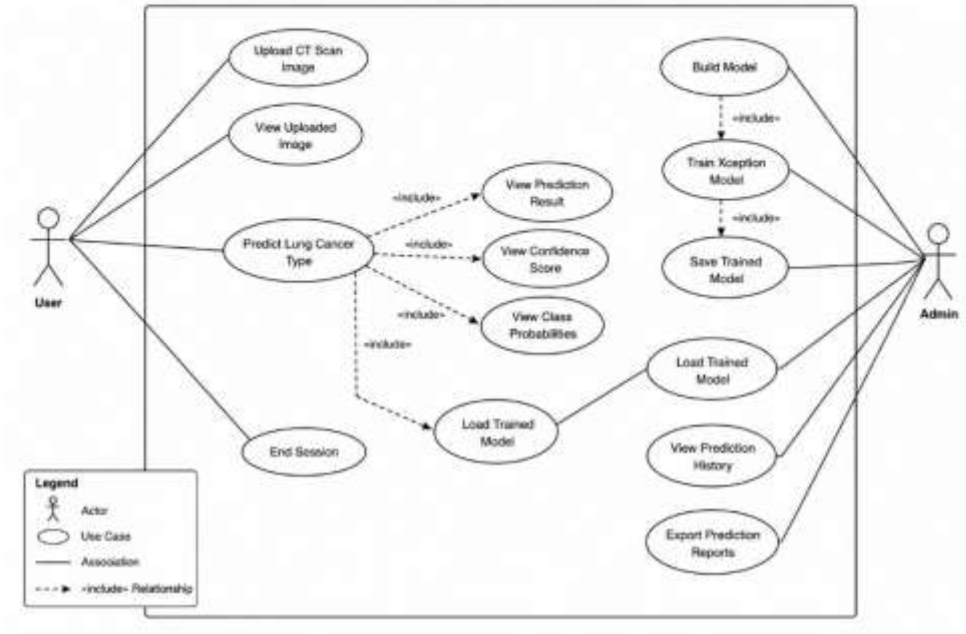


Figure 4.2: Use Case Diagram

4.1.3 Class Diagram

The Class Diagram represents the structural design of the system. It illustrates the major classes, their attributes, methods, and relationships involved in image preprocessing, model loading, prediction generation, and result visualization within the Lung Cancer Detection System.

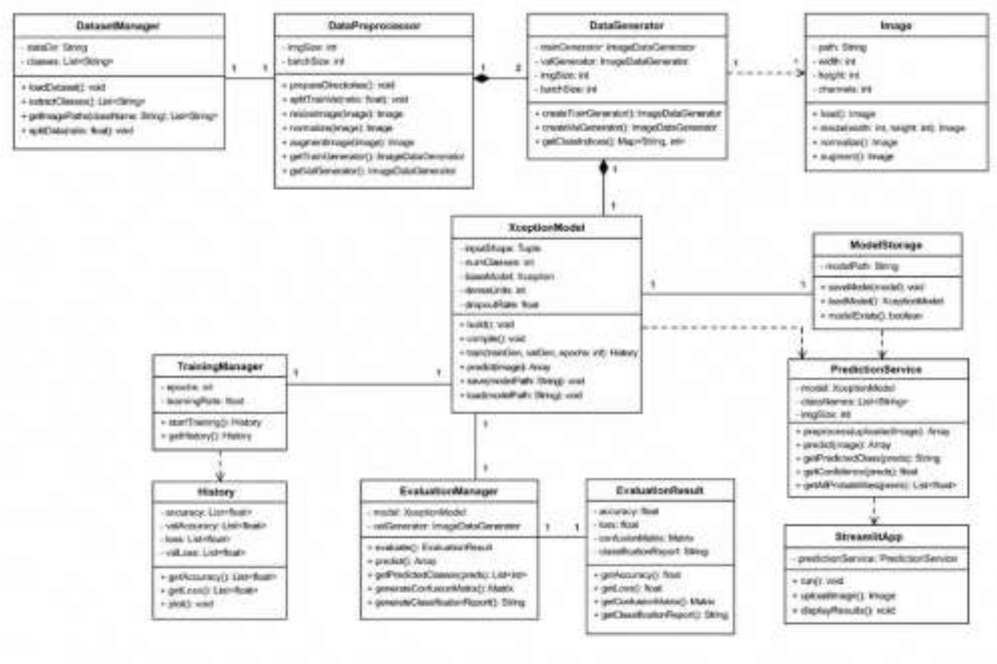


Figure 4.3: Class Diagram

4.1.4 Sequence Diagram

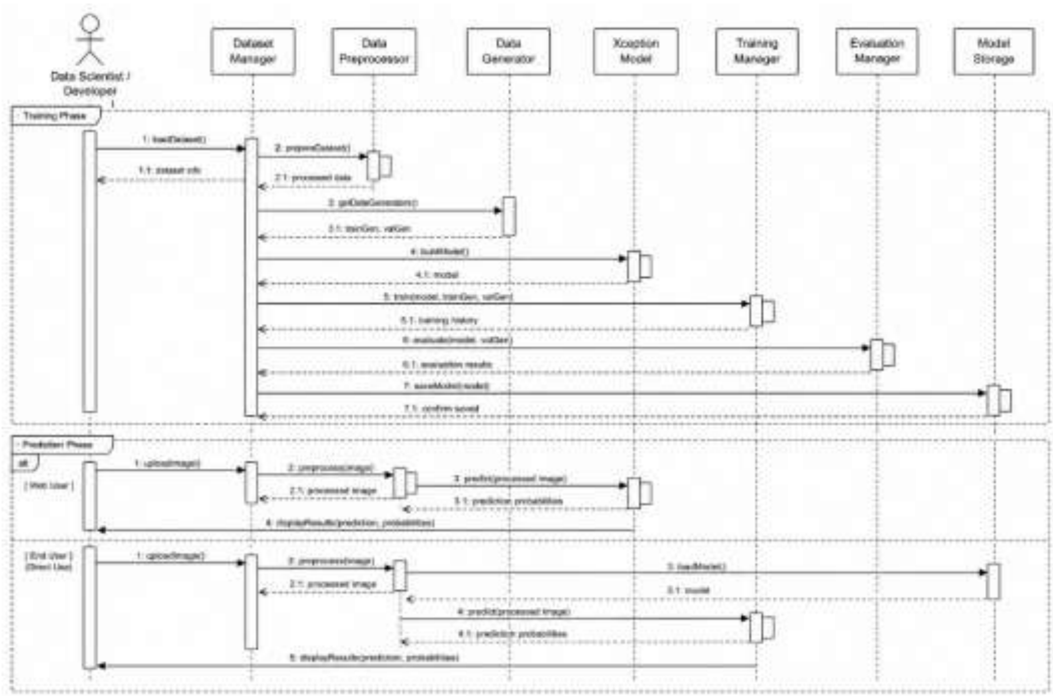


Figure 4.4: Sequence Diagram

The Sequence Diagram illustrates the chronological flow of interactions between the user, Streamlit interface, Xception model, and output components during the lung cancer classification process. It shows how data moves through the system from image upload to prediction result display.

4.2 Data Flow Diagrams

Data Flow Diagrams (DFDs) are used to represent the movement of data within the Lung Cancer Detection System. They show how CT scan images enter the system, how they are processed by the deep learning model, and how classification results are generated and returned to the user. DFDs help in understanding the functional flow and internal processing of the application.

The Data Flow Diagrams used in this project are:

- DFD Level 0
- DFD Level 1

4.2.1 DFD Level 0

The DFD Level 0 provides a high-level overview of the entire system. It represents the Lung Cancer Detection System as a single process and illustrates the interaction between the user and the system through image submission and prediction result generation.

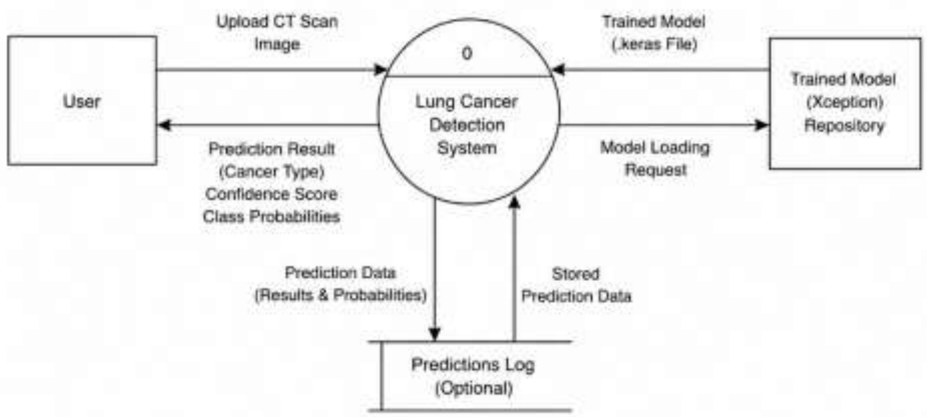


Figure 4.5: DFD Level 0 Diagram

4.2.2 DFD Level 1

The DFD Level 1 provides a detailed representation of the internal processes involved in image upload, image preprocessing, model loading, lung cancer classification using the Xception model, confidence score generation, and result visualization.

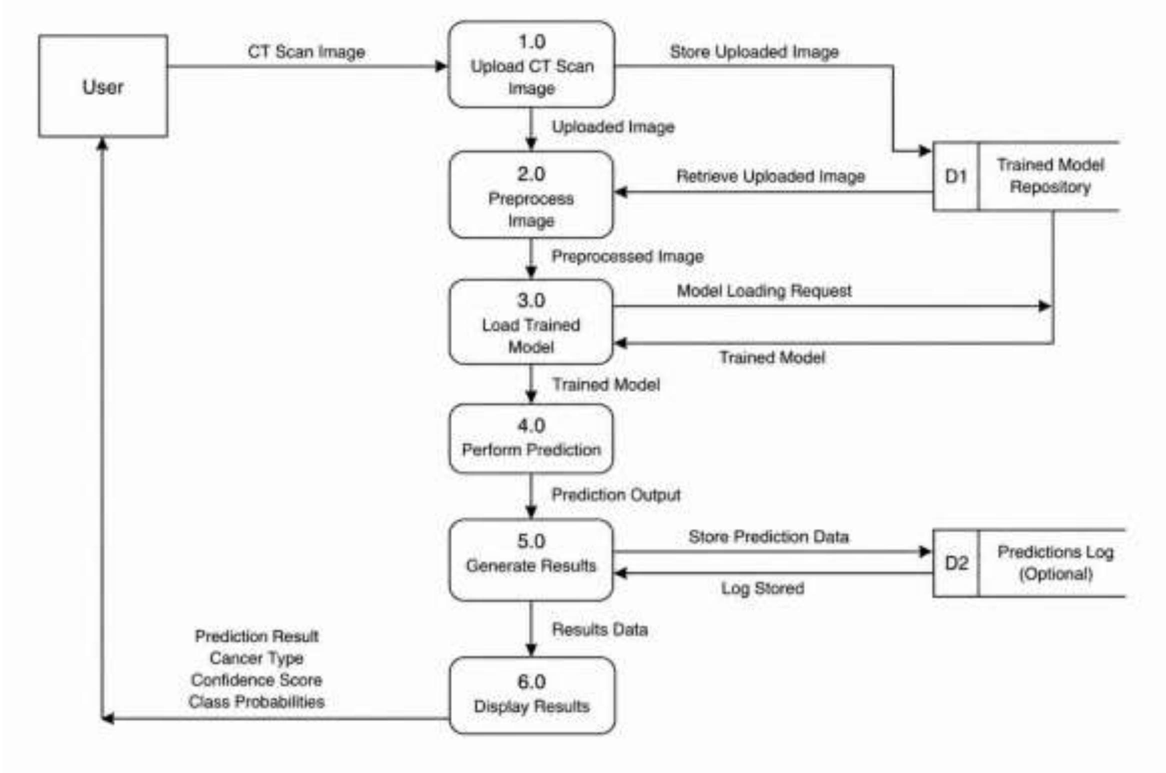


Figure 4.6: DFD Level 1 Diagram

The System Design chapter provides a comprehensive overview of the architecture and workflow of the Lung Cancer Detection System. UML diagrams describe the system structure, behavior, and interactions among users and system components, while Data Flow Diagrams illustrate the movement and processing of data within the application. Together, these diagrams provide a clear understanding of the design, functionality, and operation of the proposed lung cancer classification system.

5. IMPLEMENTATION

Implementation is the process of transforming the system design into a working software application. This phase involves the development of various modules, integration of deep learning techniques, model training, testing, and deployment of the Lung Cancer Detection System. The implementation focuses on dataset preparation, image preprocessing, model development using the Xception architecture, performance evaluation, and deployment through a Streamlit-based web application.

The system is developed using Python and various machine learning libraries including TensorFlow, Keras, Scikit-learn, NumPy, Matplotlib, and Streamlit. The trained model is capable of classifying lung CT scan images into Benign Cases, Malignant Cases, and Normal Cases.

5.1 DATASET COLLECTION

The Lung Cancer Detection System utilizes the IQ-OTHNCCD Lung Cancer Dataset as the primary source of training and testing data. The dataset contains lung CT scan images categorized into three classes:

- Benign Cases
- Malignant Cases
- Normal Cases

The dataset provides sufficient image samples for training and evaluating the deep learning model. These images are organized into separate class folders and are used for supervised learning during model development.

5.2 DATA PREPROCESSING

Data preprocessing is performed to prepare the CT scan images for model training. The preprocessing stage improves image consistency and enhances model performance.

The preprocessing operations include:

- Resizing all images to 299 × 299 pixels.

- Normalizing pixel values to a range between 0 and 1.
- Organizing images into training and validation datasets.
- Splitting the dataset into 80% training data and 20% validation data.
- Converting images into a format suitable for deep learning models.

These preprocessing steps ensure that the input images meet the requirements of the Xception architecture.

5.3 DATA AUGMENTATION

Data augmentation is applied to increase dataset diversity and improve model generalization. It helps reduce overfitting by generating modified versions of existing training images.

The augmentation techniques used include:

- Rotation
- Zooming
- Horizontal Flipping
- Rescaling

Image augmentation allows the model to learn robust features from different image variations and improves classification performance on unseen data.

5.4 MODEL DEVELOPMENT USING XCEPTION

The core of the system is based on the Xception deep learning architecture. Xception is a pre-trained Convolutional Neural Network that utilizes depthwise separable convolutions to achieve efficient feature extraction and high classification accuracy.

Transfer learning is employed by using ImageNet-trained weights. The final layers of the network are customized to suit the lung cancer classification task.

The model architecture includes:

- Pre-trained Xception base model

- Global Average Pooling Layer
- Dense Layer with ReLU activation
- Dropout Layer for regularization
- Softmax Output Layer for multi-class classification

The model is configured to classify CT scan images into Benign Cases, Malignant Cases, and Normal Cases.

5.5 MODEL TRAINING

The model is trained using the prepared dataset and optimized using the Adam optimizer. During training, the model learns to identify patterns and features associated with different lung cancer categories.

Training parameters include:

- Batch Size: 16
- Image Size: 299 × 299
- Optimizer: Adam
- Learning Rate: 0.0001
- Epochs: 8
- Loss Function: Categorical Cross-Entropy

The training process generates accuracy and loss metrics that are used to monitor model performance and convergence.

5.6 MODEL EVALUATION

After training, the model is evaluated using the validation dataset to measure its effectiveness in classifying CT scan images.

- Performance evaluation includes:
- Classification Report
- Accuracy Analysis
- Loss Analysis
- Confusion Matrix

The confusion matrix provides a visual representation of correct and incorrect classifications, while the classification report summarizes precision, recall, and F1-score values for each class.

5.7 MODEL SAVING AND DEPLOYMENT

Once training is completed, the trained Xception model is saved in Keras format for future use.

Saved Model:

- lung_cancer_xception.keras

Saving the trained model eliminates the need for retraining every time the application is executed and enables efficient deployment.

5.8 STREAMLIT-BASED USER INTERFACE

A web-based user interface is developed using Streamlit to allow users to interact with the system easily.

The Streamlit application provides the following functionalities:

- Upload lung CT scan images
- Display uploaded images
- Load the trained model automatically
- Generate predictions in real time
- Display classification results
- Display confidence scores
- Display class probabilities

The interface is designed to be simple, responsive, and user-friendly, making it accessible to both technical and non-technical users.

5.9 PREDICTION GENERATION

When a user uploads a CT scan image, the system performs the following operations:

- Accepts the uploaded image.
- Resizes the image to the required dimensions.
- Normalizes pixel values.
- Loads the trained Xception model.
- Generates prediction probabilities.
- Identifies the class with the highest probability.
- Displays the predicted category.
- Displays confidence scores and probability distribution.

The prediction process is completed within a short time, enabling rapid diagnostic assistance.

The implementation of the Lung Cancer Detection System involves dataset preparation, image preprocessing, data augmentation, model development using the Xception architecture, model training, evaluation, and deployment through a Streamlit-based web application. The integration of deep learning techniques and a user-friendly interface enables efficient classification of lung CT scan images into Benign, Malignant, and Normal categories. The implemented system provides rapid prediction results and serves as a supportive tool for lung cancer diagnosis.

6. TESTING

Testing is an essential phase in software development that ensures the developed system functions correctly, meets user requirements, and produces reliable results. The Lung Cancer Detection System was tested to verify its functionality, classification accuracy, performance, and usability. Various test cases were conducted to validate image upload functionality, CT scan image classification, prediction generation, confidence score calculation, and system responsiveness.

The primary objective of testing was to ensure that the system accurately classifies lung CT scan images into Benign Cases, Malignant Cases, and Normal Cases and provides meaningful outputs through the Streamlit interface.

6.1 Testing Objectives

The objectives of testing the Lung Cancer Detection System are:

- To verify the correct functioning of all system modules.
- To validate CT scan image upload and processing functionality.
- To evaluate the accuracy of lung cancer classification.
- To ensure proper generation of prediction results and confidence scores.
- To identify and eliminate potential errors and bugs.
- To assess system reliability and usability.
- To verify successful integration between the trained model and web application.
- To evaluate the performance of the system under different input conditions.

6.2 Types of Testing Performed

6.2.1 Functional Testing

Functional testing was performed to verify that each module of the system operates according to the specified requirements. The testing focused on image upload, image preprocessing, model prediction, result generation, and output display functionalities.

6.2.2 Integration Testing

Integration testing was conducted to ensure smooth interaction between different components, including the Streamlit interface, Xception model, image preprocessing module, prediction engine, and result visualization module.

6.2.3 Performance Testing

Performance testing evaluated the system's ability to process CT scan images efficiently and generate classification results within an acceptable response time.

6.2.4 Validation Testing

Validation testing was performed using unseen CT scan images to verify the model's capability to correctly classify Benign Cases, Malignant Cases, and Normal Cases and generalize to new data.

6.3 Test Cases

Table 6.1: Test Cases

Test Case ID	Test Description	Input	Expected Result	Actual Result	Status
TC-01	Upload Valid CT Scan Image	Valid CT scan image	Image uploaded successfully	Successful upload	Pass
TC-02	Upload Invalid File Format	PDF/Text file	System rejects unsupported file	Unsupported file rejected	Pass
TC-03	Lung Cancer Classification	CT scan image	Correct class prediction generated	Prediction generated successfully	Pass
TC-04	Benign Case Detection	Benign CT scan image	Classified as Benign Case	Correct classification obtained	Pass

TC-05	Malignant Case Detection	Malignant CT scan image	Classified as Malignant Case	Correct classification obtained	Pass
TC-06	Normal Case Detection	Normal CT scan image	Classified as Normal Case	Correct classification obtained	Pass
TC-07	Confidence Score Generation	CT scan image	Confidence score displayed	Confidence score displayed correctly	Pass
TC-08	Class Probability Display	CT scan image	Probabilities shown for all classes	Correct probabilities generated	Pass
TC-09	Multiple Image Testing	Different CT scan images	Consistent classification performance	Successful predictions generated	Pass
TC-10	Application Response Time	Image upload and prediction	Quick response and result generation	Acceptable response time	Pass

6.4 Testing Results

The testing process demonstrated that the Lung Cancer Detection System successfully performs classification of lung CT scan images. The Streamlit interface correctly accepts image uploads and displays the corresponding prediction outputs. The Xception model accurately classifies CT scan images into Benign Cases, Malignant Cases, and Normal Cases while generating confidence scores and class probabilities.

The integration between the trained model and the web application operated without significant issues. Testing with multiple CT scan images confirmed the system's ability to handle different image characteristics and provide reliable classification results.

The performance of the system was found to be satisfactory, with prediction results generated within a short response time after image upload.

6.5 System Validation

System validation was conducted using validation images from the IQ-OTHNCCD Lung Cancer Dataset as well as additional test images. The trained Xception model successfully classified CT scan images into the appropriate categories and generated accurate prediction probabilities.

The validation process confirmed:

- Correct image preprocessing workflow.
- Successful image classification functionality.
- Accurate prediction result generation.
- Proper confidence score calculation.
- Effective integration between all system components.
- Reliable performance across multiple test samples.
- Consistent classification of Benign, Malignant, and Normal cases.

The validation results demonstrate that the system can effectively assist in the preliminary classification of lung CT scan images.

. The testing phase verified the functionality, reliability, and performance of the Lung Cancer Detection System. All major modules operated successfully and produced the expected results. The Streamlit application correctly handled image uploads, the Xception model generated accurate classifications, and the system displayed prediction results along with confidence scores and class probabilities.

The test outcomes demonstrate that the system effectively classifies lung CT scan images into Benign Cases, Malignant Cases, and Normal Cases and provides users with accurate and meaningful diagnostic assistance through an easy-to-use web interface.

7. RESULTS

The Results chapter presents the outcomes obtained from the implementation of the Lung Cancer Detection System. The trained Xception model was evaluated using lung CT scan images, and the classification results were analyzed to assess the effectiveness of the proposed approach. The system successfully classified CT scan images into Benign Cases, Malignant Cases, and Normal Cases while generating confidence scores and class probabilities through a user-friendly Streamlit interface.

7.1 Present Results Obtained

The Lung Cancer Detection System was successfully developed, trained, and deployed. The Xception model demonstrated the ability to accurately classify lung CT scan images into their respective categories. During testing, the model generated prediction results along with confidence scores indicating the certainty of each classification.

The integration of the trained model with the Streamlit web application enabled users to upload CT scan images and obtain classification results in real time. The system performed effectively across various test images and demonstrated reliable image classification capabilities.

The major outcomes achieved are:

- Successful training of the Xception model using the IQ-OTHNCCD Lung Cancer Dataset.
- Accurate classification of CT scan images into Benign Cases, Malignant Cases, and Normal Cases.
- Generation of confidence scores and class probabilities for each prediction.
- Development of a user-friendly Streamlit-based web application.
- Real-time prediction and result visualization.
- Effective integration of deep learning and medical image analysis techniques.
- Reliable performance across different CT scan image samples.
- Automated support for preliminary lung cancer diagnosis.

7.2 Screenshots

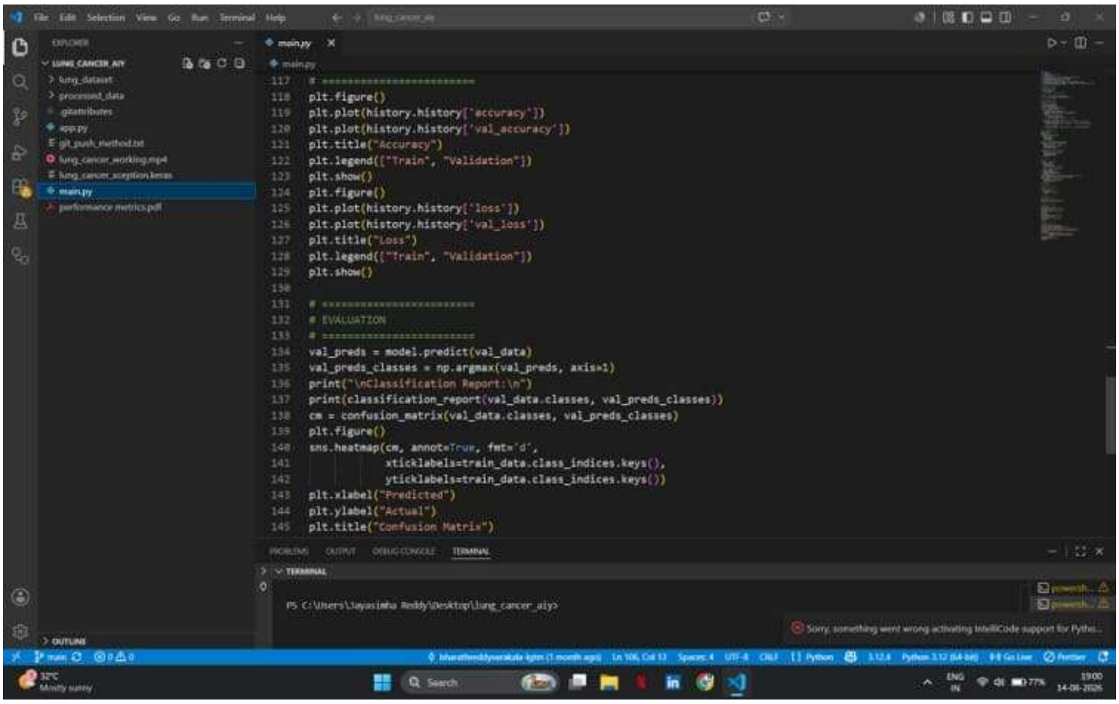


Figure 7.1: Successful compilation of code

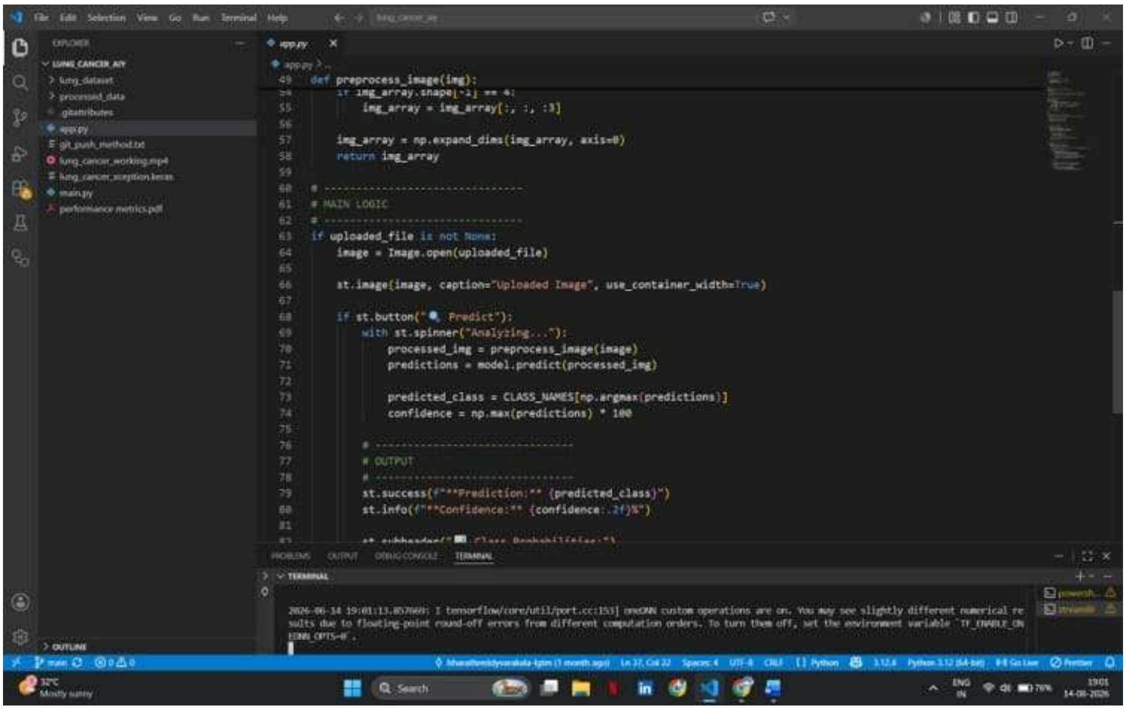


Figure 7.2: Successful compilation of code

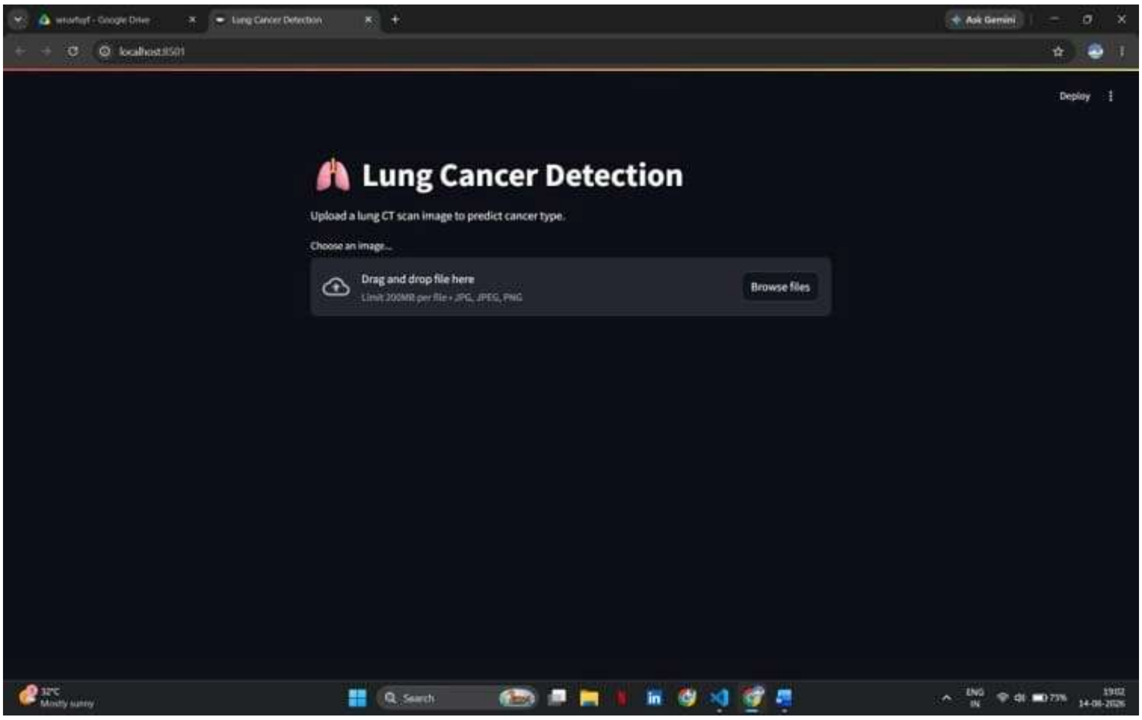


Figure 7.3: Streamlit web application landing page

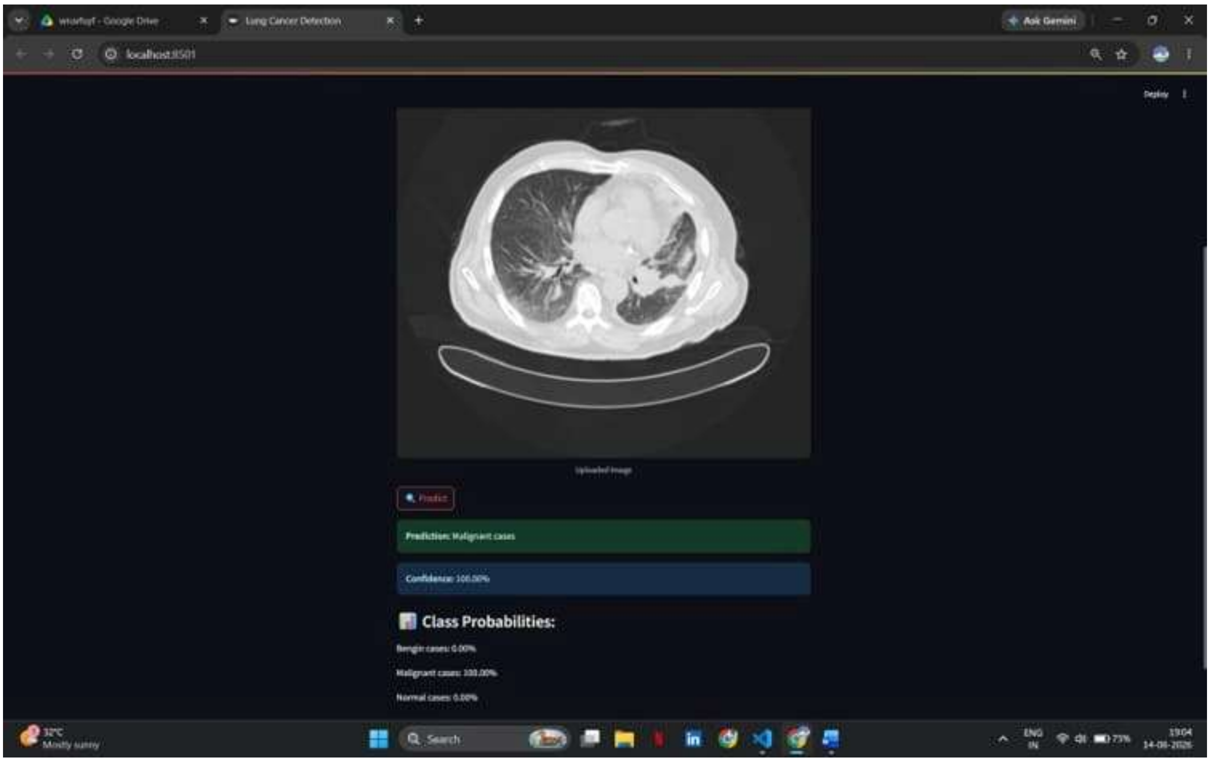


Figure 7.4: Prediction results for Test Image - 01

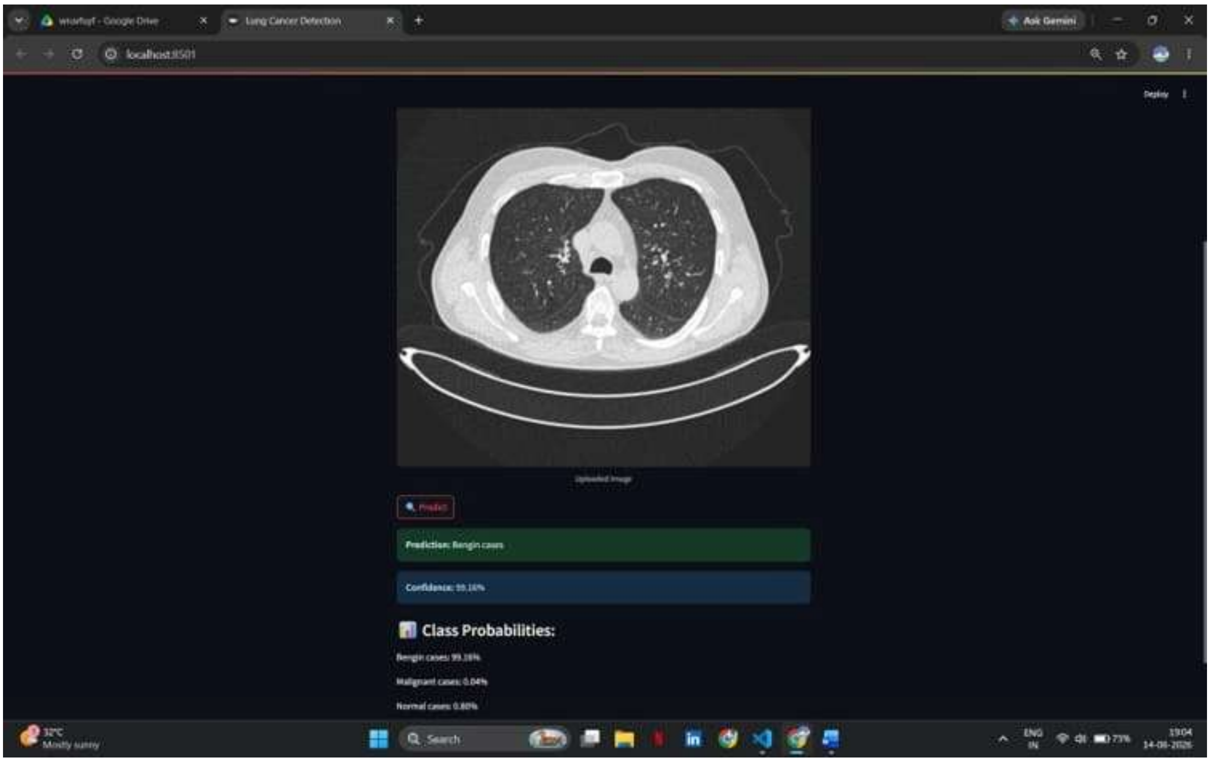


Figure 7.5: Prediction results for Test Image – 02

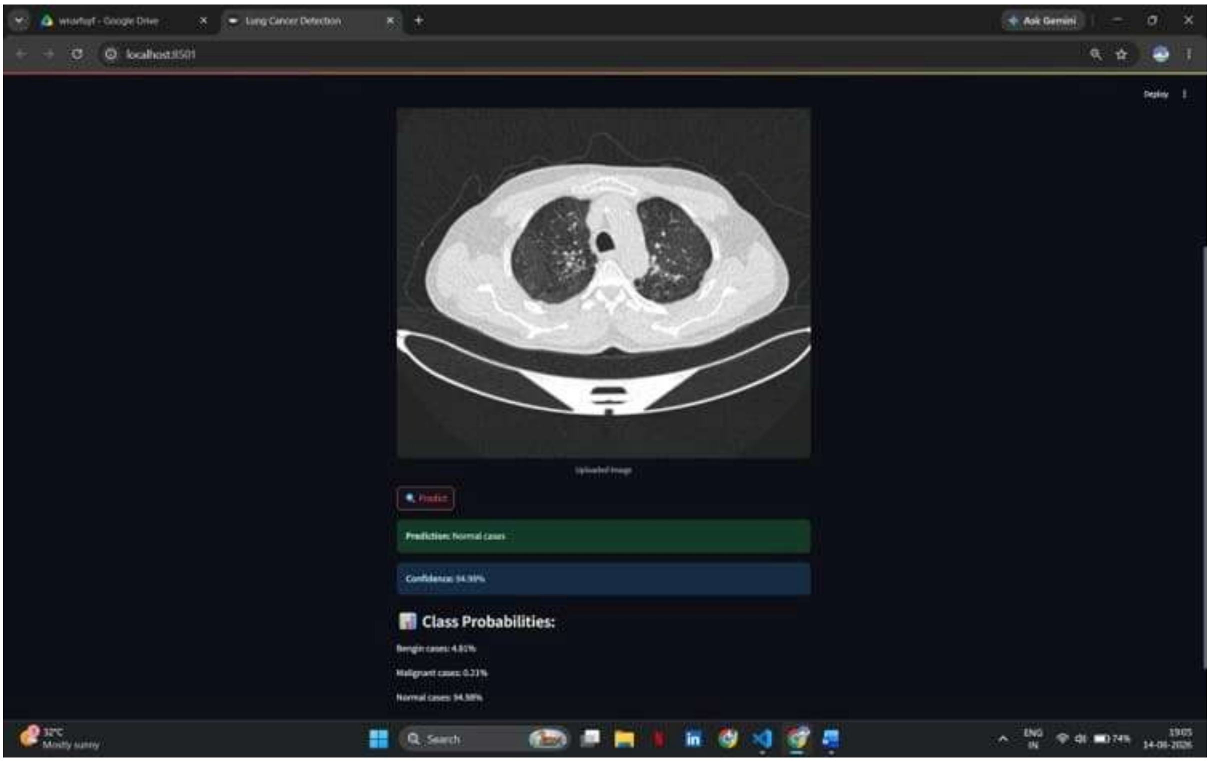


Figure 7.5: Prediction results for Test Image – 03

7.3 Performance Analysis

The performance of the Lung Cancer Detection System was evaluated based on its classification capability, prediction speed, and usability. The Xception architecture provided efficient image classification while maintaining a fast response time suitable for practical applications.

The trained model successfully classified CT scan images despite variations in image characteristics and medical imaging conditions. The use of transfer learning and data augmentation techniques during training contributed to improved model generalization and robustness.

Key performance observations include:

- Fast image preprocessing and prediction generation.
- Accurate classification of lung CT scan images.
- Reliable identification of Benign, Malignant, and Normal cases.
- Consistent performance across multiple test samples.
- Efficient deployment through the Streamlit platform.
- Reduced manual effort in preliminary diagnostic analysis.
- User-friendly interface for prediction and result visualization.
- Effective utilization of deep learning for medical image classification.

The combination of the Xception architecture and the IQ-OTHNCCD Lung Cancer Dataset enabled the system to achieve effective classification performance suitable for computer-aided lung cancer diagnosis applications.

7.4 Interpretation of Results

The obtained results indicate that the proposed Lung Cancer Detection System is capable of effectively classifying lung CT scan images into Benign Cases, Malignant Cases, and Normal Cases. The Xception model successfully learned discriminative features from the training dataset and applied this knowledge to classify unseen CT scan images.

The generated confidence scores and class probabilities provide additional information regarding the reliability of each prediction. These outputs help users better understand the classification results produced by the system.

The results demonstrate that deep learning-based medical image classification can significantly improve the efficiency of preliminary diagnostic analysis compared to traditional manual approaches. The successful deployment of the model through a Streamlit interface further confirms the practical applicability of the proposed system for healthcare, medical research, and educational purposes.

The overall performance of the system suggests that deep learning can serve as a valuable tool for assisting healthcare professionals in the early detection and classification of lung cancer.

The results obtained from the Lung Cancer Detection System confirm the effectiveness of the Xception deep learning architecture for lung CT scan image classification. The system successfully achieved its objectives by providing accurate classification, real-time prediction, and an accessible user interface. The integration of deep learning techniques with medical image analysis enabled reliable identification of Benign Cases, Malignant Cases, and Normal Cases. These outcomes demonstrate the potential of artificial intelligence in supporting early lung cancer diagnosis and improving healthcare decision-making processes.

\

CONCLUSION

The Lung Cancer Detection System was successfully developed using deep learning techniques to classify lung CT scan images into Benign Cases, Malignant Cases, and Normal Cases. The system utilizes the Xception transfer learning architecture, which effectively extracts important features from CT scan images and performs accurate image classification.

The project demonstrates the potential of artificial intelligence in assisting medical diagnosis by automating the analysis of lung CT scan images. Through image preprocessing, data augmentation, model training, and evaluation, the developed system achieved reliable classification performance and generated meaningful prediction results. The integration of the trained model with a Streamlit-based web application further enhanced the usability of the system by providing a simple and interactive platform for users.

The developed application enables users to upload CT scan images and receive real-time predictions along with confidence scores and class probabilities. This reduces the time required for preliminary analysis and can assist healthcare professionals in making informed diagnostic decisions.

Overall, the project successfully achieved its objectives by developing an efficient, accurate, and user-friendly Lung Cancer Detection System. The results obtained demonstrate the effectiveness of deep learning-based medical image classification and highlight its potential role in supporting early lung cancer diagnosis and improving healthcare outcomes.

FUTURE ENHANCEMENTS

Although the proposed Lung Cancer Detection System provides accurate and reliable classification results, several improvements can be made to further enhance its functionality, performance, and practical applicability.

Future enhancements of the system may include:

- Expanding the dataset with a larger number of CT scan images to improve model accuracy and generalization.
- Incorporating additional lung disease categories and cancer stages for more detailed diagnosis.
- Implementing advanced deep learning architectures and ensemble models to achieve higher classification performance.
- Integrating explainable AI techniques to provide visual explanations and improve the interpretability of predictions.
- Developing a cloud-based deployment platform to enable remote access and large-scale usage.
- Integrating the system with hospital information systems and electronic health records for clinical applications.
- Supporting real-time processing of medical imaging data from healthcare devices and diagnostic equipment.
- Developing a mobile application to provide convenient access to prediction services for healthcare professionals.
- Enhancing the user interface with advanced visualization and reporting features.
- Conducting extensive clinical validation using larger and more diverse medical datasets to improve reliability in real-world healthcare environments.

These enhancements can further improve the accuracy, scalability, accessibility, and practical usefulness of the Lung Cancer Detection System, making it a more effective tool for assisting healthcare professionals in the early detection and diagnosis of lung cancer.

REFERENCES

[1] A. Kumar and R. Sharma, “Lung Cancer Detection Using Deep Learning Techniques,” *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 8, pp. 245–251, 2020.

[2] M. Al Mamun and S. Islam, “Automated Lung Cancer Detection from CT Images Using Transfer Learning,” *International Journal of Computer Applications*, vol. 174, no. 22, pp. 12–18, 2021.

[3] H. Wang and Y. Zhang, “Deep Learning-Based Computer-Aided Diagnosis System for Lung Cancer Detection,” *IEEE Access*, vol. 8, pp. 125432–125441, 2020.

[4] S. Gupta and P. Verma, “Lung Cancer Classification Using Xception Network and Medical Imaging,” *International Journal of Intelligent Systems and Applications*, vol. 14, no. 3, pp. 34–42, 2022.

[5] K. Rajesh and M. Prakash, “Transfer Learning Approaches for Medical Image Classification,” *Journal of Artificial Intelligence and Data Science*, vol. 6, no. 2, pp. 89–98, 2021.

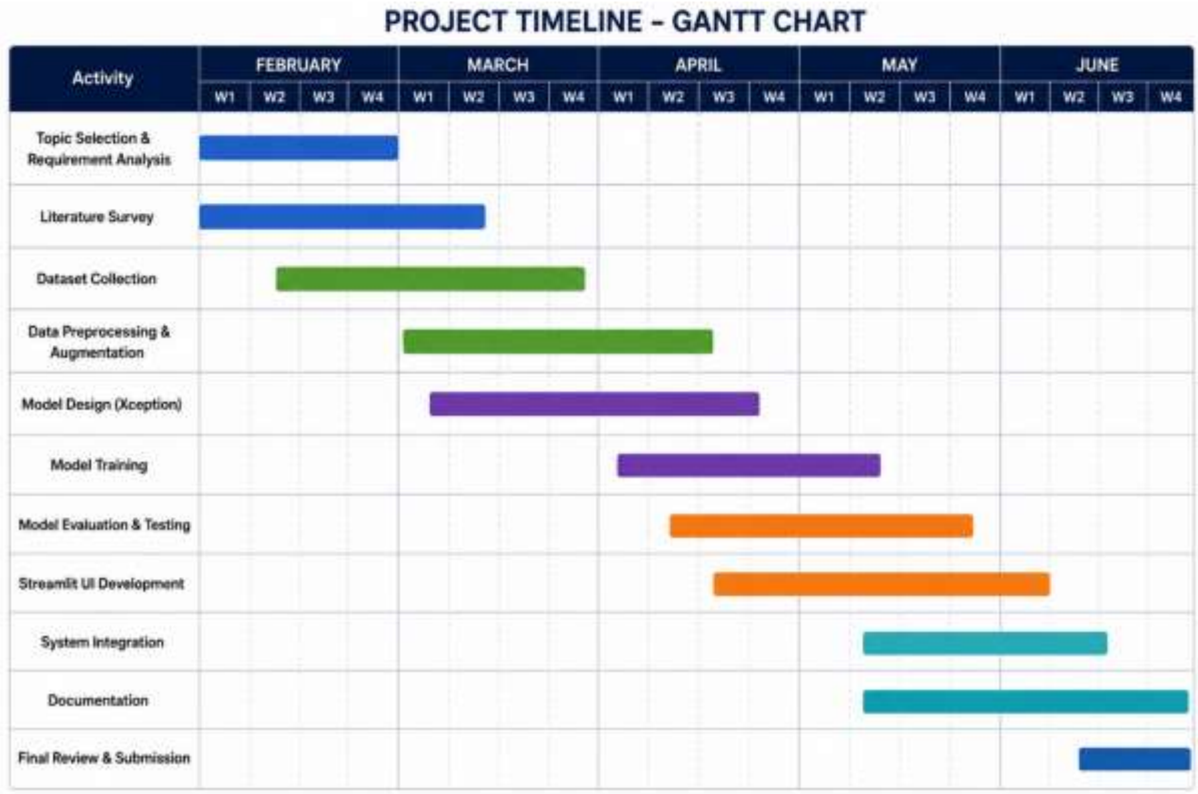
[6] J. Singh and R. Kaur, “Artificial Intelligence in Lung Cancer Diagnosis: A Review,” *International Journal of Medical Informatics*, vol. 175, pp. 105021, 2023.

[7] T. Ahmed and M. Hassan, “CT Scan-Based Lung Disease Classification Using Deep Neural Networks,” *Procedia Computer Science*, vol. 192, pp. 1842–1851, 2022.

APPENDIX

A. GANTT CHART

Project Progress Timeline (14 Weeks)



B. SAMPLE CODE

1. Dataset loading and preparation

```
data_dir = "lung_dataset/The IQ-OTHNCCD lung cancer dataset"
classes = [
    cls for cls in os.listdir(data_dir)
    if os.path.isdir(os.path.join(data_dir, cls))
]
print("Classes:", classes)
```

2. Data augmentation and image processing

```
train_gen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=15,
    zoom_range=0.2,
    horizontal_flip=True
)
val_gen = ImageDataGenerator(rescale=1./255)
```

3. Xception model construction

```
base_model = Xception(
    weights='imagenet',
    include_top=False,
    input_shape=(299, 299, 3)
)
for layer in base_model.layers[:-20]:
    layer.trainable = False
```

4. Classification layer and model compilation

```
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(512, activation='relu')(x)
x = Dropout(0.5)(x)
```

```
output = Dense(
    len(train_data.class_indices),
    activation='softmax'
)(x)
model = Model(
    inputs=base_model.input,
    outputs=output
)
model.compile(
    optimizer=Adam(learning_rate=1e-4),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

5. **Prediction using streamlit application**

```
processed_img = preprocess_image(image)
predictions = model.predict(processed_img)
predicted_class = CLASS_NAMES[
    np.argmax(predictions)
]
confidence = np.max(predictions) * 100
st.success(
    f"Prediction: {predicted_class}"
)
st.info(
    f"Confidence: {confidence:.2f}%"
)
```

C. PROJECT MAPPING

QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)			
Course Name:	MINI PROJECT	**AY**	2025-26
Course Code:	7PW663ML	**Semester**	VI
Name of the Guide:	Ms. P LAKSHMI	**Class**	AI&ML - A

PROJECT TITLE
LUNG CANCER DETECTION USING XCEPTION LEARNING

STUDENTS

S. No	HT No	Name of the Student
1.	160723748009	Mohammed Inabah Khan
2.	160723748018	Bannenola Akhil
3.	160723748041	Guttula Yeshwanth

Course Outcome: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	POs, PSOs Mapped
7PW663ML.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	**Apply**	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.2	Analyze and formulate the problem, and design solution of the selected project.	**Analyze**	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.3	Evaluate different solutions based on economic and technical feasibility.	**Evaluate**	PO3, PO4, PO5, PO6, PO7, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.4	Effectively plan, develop and deploy a project and confidently perform all aspects of project management.	**Create**	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3
7PW663ML.5	Demonstrate effective written and oral technical communication skills.	**Create**	PO2, PO5, PO7, PO9, PO10, PO12, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
7PW663ML. 1	3	2	2	-	-	3	-	2	-	-	2	-	3	3	3
7PW663ML. 2	3	3	2	3	2	2	-	2	-	-	2	-	3	3	3
7PW663ML. 3	-	-	3	3	2	2	2	2	-	-	3	-	3	2	2

7PW663ML. 4	-	-	3	3	3	2	2	3	3	3	3	2	3	3	3
7PW663ML. 5	-	2	-	-	2	-	2	2	3	3	2	2	2	2	2

CO-PO-PSO MAPPING (WITH PERFORMANCE INDICATORS)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS0 1	PSO 2	PSO 3
7PW663ML.1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2	-	-	6.1.1 6.2.1 6.3.1 6.3.2 6.4.1	-	8.2.1 8.2.2 8.2.3	-	-	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	-	**High**	**High**	**High**
7PW663ML.2	1.1.2 1.2.1 1.3.1 1.4.1	2.1.1 2.1.2 2.1.3 2.2.3 2.3.1 2.4.1 2.4.3 2.4.4	3.1.1 3.1.3 3.1.4 3.1.5 3.1.6 3.2.1 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.1 4.1.2 4.1.3 4.2.1 4.3.1 4.3.2 4.3.3 4.3.4	-	6.1.1 6.3.1 6.3.2 6.4.2	-	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	-	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**High**	**High**
7PW663ML.3	-	2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**Mod**	**Mod**
7PW663ML.4	-	2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**High**	**High**	**High**
7PW663ML.5	-	2.2.1 2.2.2 2.2.3 2.2.4	-	-	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	-	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**Mod**	**Mod**	**Mod**

MINI PROJECT SPECIFIC LEARNING OUTCOMES

LO#	Outcome Description
1	Understand and apply deep learning techniques for medical image classification using CT scan images.
2	Perform data preprocessing, augmentation, and dataset preparation for machine learning model development.
3	Develop Design, train, evaluate, and optimize an Xception-based transfer learning model for lung cancer detection.
4	Develop and deploy a user-friendly web application using Streamlit for real-time prediction and result visualization.
5	Analyze prediction results and demonstrate the practical application of Artificial Intelligence in healthcare diagnostics.

PO -WK MAPPING:

PO#	PO Description	Mapped WKs	PO#	PO Description	Mapped WKs
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Environment and sustainability	WK5, WK7
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Ethics	WK7, WK9
PO3	Design / Development of Solutions	WK1, WK2, WK3, WK4, WK5, WK6, WK7	PO9	Individual & Team Work	WK9
PO4	Conduct Investigations of complex problems	WK2, WK3, WK4, WK8	PO10	Communication	WK7, WK9
PO5	Engineering Tool usage	WK2, WK6	PO11	Project Management & Finance	WK5, WK6, WK7
PO6	The Engineer and society	WK1, WK5, WK7	PO12	Life-long Learning	WK8

SDG MAPPING:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	3.1.1	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.5.1	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	17.1.1

6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

QUALITY ANALYSIS:

MINI PROJECT COLLABORATION

(Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other
	✓		✓				

MINI PROJECT ACHIEVEMENTS:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes

S. No	**Description**	**Details**
1.	Prototype development	Developed a functional AI-based lung cancer detection web application.
2.	Technology based solution developed	Implemented Deep Learning and Transfer Learning techniques for automated CT scan classification.
3.	Publication	Nil
4.	Start-up incubation	Nil
5.	Competition prizes	Nil
6.	Community / Industry / HEI service	Supports healthcare awareness and AI-assisted diagnostic research.
7.	SDG Indicators	SDG 3, SDG 4, and SDG 9
8.	Grants / Financial Aid	Nil
9.	Multi-disciplinary	Combines Artificial Intelligence, Machine Learning, Medical Imaging, and Healthcare.
10.	Internship / Employment	Provides practical skills relevant to AI, ML, Data Science, and Healthcare Analytics careers.
11.	Self-employment skill	Enhances capability to develop and deploy AI-based software solutions independently.
12.	Use of real-time data	Uses real medical CT scan image data from the IQ-OTHNCCD Lung Cancer Dataset.

Signature of the Guide **Signature of the Project Coordinator**

(QR CODE with all project docs in Gdrive)